

Blinkit Sales & Profitability Analysis

Project Documentation

A comprehensive data analysis workflow uncovering key insights into sales performance, revenue optimization, and product margin trends.

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Date: July 2026

Dataset: Blinkit Grocery Sales (Kaggle)

Tools & Technologies

 **Python**— Core programming language

 **Pandas & NumPy**— Data manipulation

 **Matplotlib**— Data visualization

 **Google Colab**— Interactive environment

Project Showcase

Business Problem

- Revenue and profitability vary across product categories, making it difficult to identify the most valuable segments.
- Discounts and promotional offers can impact both sales performance and profit margins.
- Customer ratings and demand patterns influence product performance and purchasing behavior.
- Inefficient inventory planning can lead to stock shortages or excess inventory.
- Data-driven insights are needed to optimize pricing, inventory management, and overall business performance.

Business Objectives

- Analyze revenue performance across product categories and identify top-performing products.
- Evaluate profit trends and assess the impact of discounts on revenue and profitability.
- Analyze customer ratings and purchasing behavior to understand sales patterns.
- Assess inventory availability and operational performance to identify improvement opportunities.
- Generate actionable insights and recommendations to improve sales, profitability, and business efficiency.

Dataset Overview

01. DATASET INFORMATION

- Dataset Name: Blinkit Grocery Sales Dataset
- Source: Kaggle
- File Format: CSV

02. DATASET SIZE

- Total Records: 13,000
- Total Columns: 25

03. KEY DATA CATEGORIES

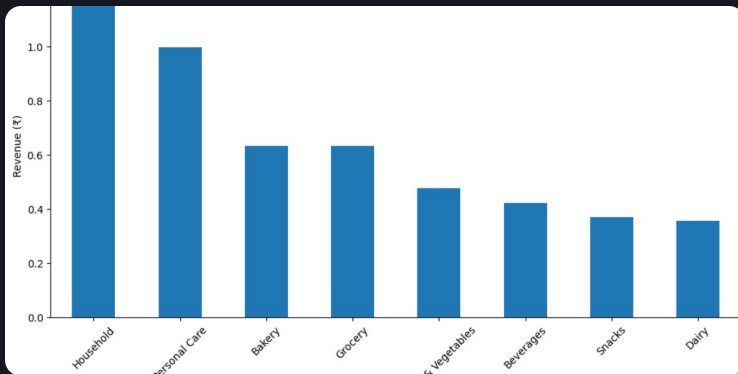
- Product Information & Pricing Details
- Sales Performance & Customer Ratings
- Delivery Details & Inventory Information

04. PURPOSE OF THE DATASET

- Analyze product sales performance & profitability
- Study customer purchasing behavior
- Generate data-driven business insights

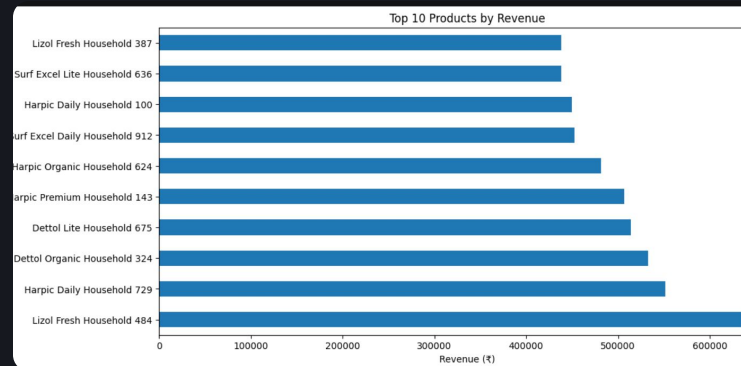
Revenue & Product Performance

01. REVENUE BY CATEGORY



Insight: Household and Personal Care categories generated the highest revenue, indicating strong customer demand.

02. TOP 10 PRODUCTS BY REVENUE

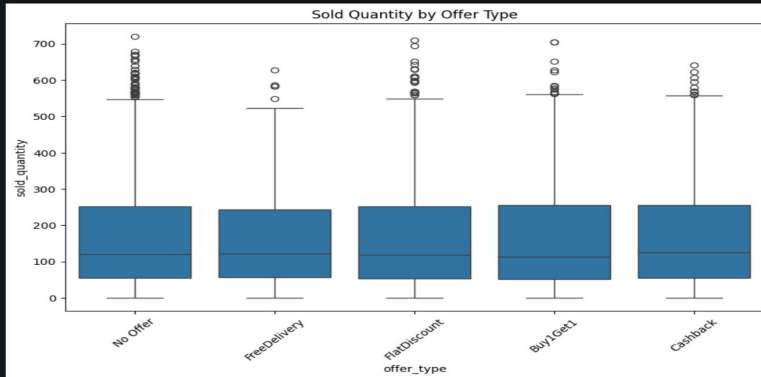


Insight: A small group of products contributed a significant share of total revenue, highlighting key revenue drivers.

Analysis & Visualizations (Part 1)

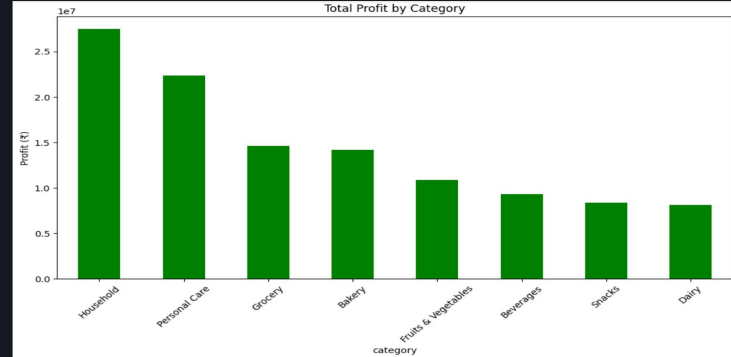
Quantity & Profit Analysis

01. SOLD QUANTITY BY OFFER TYPE



Insight: Promotional offers increased sales volume, but their impact on profitability required further evaluation.

02. TOTAL PROFIT BY CATEGORY

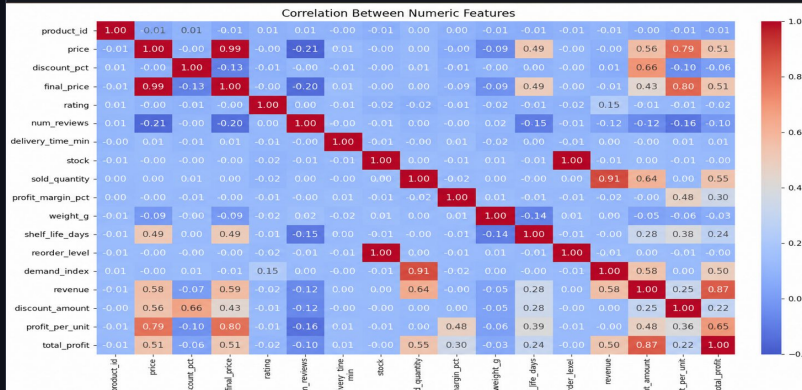


Insight: High-revenue categories also delivered the strongest profit contribution, making them priority segments for growth.

Analysis & Visualizations (Part 2)

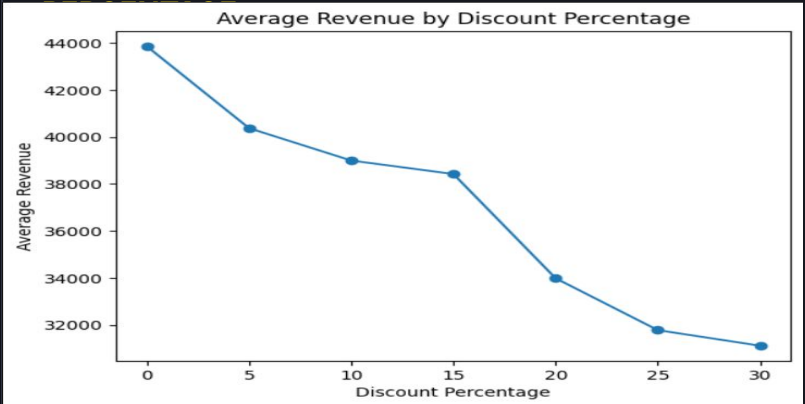
Correlation & Revenue Impact

01. CORRELATION BETWEEN NUMERIC FEATURES



Insight: Revenue showed a strong positive correlation with sold quantity and total profit, indicating that higher sales volumes contributed directly to business profitability.

02. AVERAGE REVENUE BY DISCOUNT



Insight: Average revenue decreased as discount percentages increased, suggesting that excessive discounts did not improve revenue performance.

Executive Summary

Key Insights

01. HIGH-PERFORMING CATEGORIES

Household and Personal Care categories generated the highest revenue and profit.

02. REVENUE CONCENTRATION

A small group of products generated a significant share of the overall revenue.

03. IMPACT OF DISCOUNTS

Higher discount percentages were associated with lower average revenue, suggesting that excessive discounts may reduce profitability.

04. REVENUE DRIVERS

Higher sold quantities were strongly associated with increased revenue and total profit.

05. PURCHASING BEHAVIOR

Ratings alone did not determine sales, suggesting pricing and demand heavily influence purchasing decisions.

06. BUSINESS OPPORTUNITY

Data analysis identified opportunities to optimize pricing strategies, inventory planning, and overall business performance.

Strategic Recommendations

QUICK WINS (0–3 Months)

- Reduce excessive discount campaigns and focus on targeted promotions.
- Increase inventory availability for Household and Personal Care products.
- Maintain sufficient stock for top-selling products to prevent lost sales.
- Allocate inventory based on demand patterns and sold quantity trends.

LONG-TERM (3–12 Months)

- Implement demand forecasting using historical sales and demand data.
- Develop dynamic pricing strategies to balance revenue growth and profitability.
- Create performance dashboards to monitor sales, profit, inventory, and demand metrics.
- Improve inventory planning using demand trends and reorder-level analysis.

Conclusion

Project Wrap-Up

01. PROJECT SUMMARY

Successfully analyzed Blinkit's grocery sales dataset using Python and exploratory data analysis techniques.

02. KEY FINDINGS

Identified major revenue drivers, profitability trends, and the impact of discount strategies on business performance.

03. BUSINESS VALUE

Generated actionable insights to support pricing optimization, inventory management, and strategic decision-making.

04. SKILLS APPLIED

Demonstrated practical skills in Python, Pandas, NumPy, Matplotlib, Exploratory Data Analysis (EDA), and Business Analytics.

05. FINAL OUTCOME

Transformed raw sales data into actionable business insights, enabling data-driven decision-making and improving business performance.